

**NMAM INSTITUTE OF TECHNOLOGY, NITTE***(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CCE) (Credit System) Degree Examinations**

January 2023

**20CC502 - CRYPTOGRAPHY AND NETWORK SECURITY**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer Five full questions choosing One full question from each Unit.*

| Unit – I   |                                                                                                                                                                                    | Marks | BT* | CO* | PO* |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a)      | List and discuss various types of cryptanalytic attacks based on the amount of information known to the cryptanalyst.                                                              | 6     | L*2 | 1   | 1   |
| b)         | What are the rules to encrypt the message in Play fair cipher? Consider the key as "SCIENCE". Encrypt the message "AUTONOMOUS" using Play fair cipher.                             | 8     | L3  | 1   | 2   |
| c)         | Discuss single round of DES Algorithm.                                                                                                                                             | 6     | L3  | 1   | 1   |
| 2. a)      | How mono alphabetic cipher is different from Caesar cipher? Discuss the attacks on Mono alphabetic cipher.                                                                         | 6     | L3  | 1   | 1   |
| b)         | Discuss encryption and decryption process in Hill Cipher. Encrypt the message "letusm" using the Hill Cipher with the key<br>$\begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$        |       |     |     |     |
|            | Show all calculations and result. Show how to decrypt first two characters.                                                                                                        | 8     | L4  | 1   | 2   |
| c)         | Discuss Block cipher design principle.                                                                                                                                             | 6     | L3  | 1   | 1   |
| Unit – II  |                                                                                                                                                                                    |       |     |     |     |
| 3. a)      | Discuss public key crypto systems with encryption with public key and encryption with private key.                                                                                 | 10    | L2  | 2   | 1   |
| b)         | Design Diffie-Hellman key exchange Algorithm. How man-in-the-middle attack can be done for the Algorithm?                                                                          | 10    | L3  | 2   | 1   |
| 4. a)      | Discuss RSA Algorithm along with suitable example.                                                                                                                                 | 10    | L3  | 2   | 1,2 |
| b)         | Discuss Elliptic-curve Cryptography method. Comment on security of this scheme.                                                                                                    | 10    | L3  | 2   | 2   |
| Unit – III |                                                                                                                                                                                    |       |     |     |     |
| 5. a)      | With respect to symmetric key distribution using symmetric encryption, discuss the following.<br>i) Decentralized key distribution<br>ii) Control vector encryption and decryption | 12    | L3  | 3   | 1   |
| b)         | Discuss the NIST model for Electronic user authentication.                                                                                                                         | 8     | L3  | 3   | 1   |
| 6. a)      | With respect to distribution of public keys, discuss the following.<br>i) Public key authority ii) Public key certificate                                                          | 12    | L3  | 3   | 1   |
| b)         | List four requirements for Kerberos. Discuss Ticket-Granting-Server scheme.                                                                                                        | 8     | L2  | 3   | 1   |
| Unit – IV  |                                                                                                                                                                                    |       |     |     |     |
| 7. a)      | What are the security risks of wireless networks compared to wired networks? Discuss the various threats to wireless networks.                                                     | 10    | L3  | 4   | 1   |
| b)         | Discuss Transport Layer security architecture.                                                                                                                                     | 10    | L2  | 4   | 1   |
| 8. a)      | Discuss IEEE 802.11 protocol stack. List the various services of IEEE 802.11                                                                                                       | 10    | L3  | 4   | 1   |
| b)         | What is Secure Shell? Discuss SSH Protocol stack.                                                                                                                                  | 10    | L3  | 4   | 1   |
| Unit – V   |                                                                                                                                                                                    |       |     |     |     |
| 9. a)      | Discuss various services provided by Pretty Good Privacy.                                                                                                                          | 10    | L2  | 5   | 1   |
| b)         | What are the benefits of IPSec and IPSec services?                                                                                                                                 | 10    | L2  | 5   | 1   |
| 10. a)     | Discuss DKIM functional flow in detail.                                                                                                                                            | 10    | L3  | 5   | 1   |
| b)         | Explain ESI packet format with various fields.                                                                                                                                     | 10    | L3  | 5   | 1   |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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